Week 6 — Forecasting Field Guide

The patient funnel: a simple model with documented assumptions

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Reference handout · keep for Homework 2 and the term project · Monday, October 19, 2026

This guide covers the vocabulary, the formulas and the templates in the order a forecast is built. The model is deliberately simple: a few multiplications per year, each input written down with a range, a source and a reason. Worked numbers are from tonight's synthetic LYMBREXA / DLBCL case; the same model is in Week_06_LYMBREXA_Forecast_Model.xlsx and the notebook. Sources are peer-reviewed papers, standard texts, government agencies and the IQVIA papers in the course library (References at the end).

The principle

A forecast is a chain of assumptions multiplied together. Keep the chain short, write every link down, and show a range. A simple model that everyone can challenge beats a complex one nobody can follow.

Part 1 · What a forecast is for

	Budget forecast	Long-range plan (LRP)
Horizon	next 2–3 years	5–15 years, through loss of exclusivity
Grain	monthly; by region and account	annual
Refresh	quarterly	annually, and after major events (trial readout, label)
Costly error	a miss in the next 12 months: supply, cash, sales targets	the size and timing of the peak: investment decisions
Owner	brand lead + finance	portfolio strategy + finance + R&D

Build one model. The budget is the LRP's first two or three years, spread by month.

Function	Uses the forecast for
Finance	the budget; guidance to investors
Supply / manufacturing	how many vials, by month; plant capacity years ahead
Sales operations	targets and incentive pay
Marketing	how much to spend, and where
Market access	rebates and contract volumes
Medical affairs	team size; evidence plans
R&D / business development	which indication next; what a deal is worth
Executive team	the plan of record

Part 2 · Choosing the model shape

Incidence model. Starts from new cases per year. Fits oncology lines of therapy, acute care and vaccines. Tonight: DLBCL.

Prevalence model. Starts from people living with the disease. Fits chronic therapy: diabetes, psoriasis, HIV. Tonight: GLUCOREN in type 2 diabetes. Prevalence ≈ incidence × how long the disease lasts.

Flow model. Patients move through stages with a delay (here: one year from line to line). Use it when timing matters — launches, new labels, patent expiry.

Stock model. Patients this year × rates × shares; each year stands alone. Fine for a stable market.

Funnel	Starts from	Usual bottleneck
Oncology line of therapy (incidence, flow)	new cases per year	progression rates between lines
Chronic disease (prevalence, stock)	people living with it	treated rate and brand share
Rare disease (prevalence, patient finding)	people living with it, mostly undiagnosed	diagnosis and referral

Part 3 · The LYMBREXA model in seven lines

1. 1L patients(y) = new cases(y) × diagnosed × treated in 1L
2. 2L starts(y) = 1L patients(y−1) × share progressing to 2L
3. 3L+ starts(y) = 2L starts(y−1) × share progressing to 3L × later-lines multiplier
4. brand share(y) = peak share × uptake(year since launch) × coverage × LOE retention
5. new patients(y) = 3L+ starts × 3L+ share + 2L starts × 2L share
6. patients on therapy(y) = new patients(y) × months on therapy ÷ 12
7. net revenue(y) = new patients(y) × months on therapy × compliance × list price per month × gross-to-net

Budget: take the 2027 and 2028 values and spread them by month — a straight-line ramp in the launch year (month m gets m/78 of the year), equal months after. That phasing is itself an assumption (F1, F2).

Worked example — 2032, base case

3L+ starts 3,931 × 15% × 95% = 560 new 3L+ patients; 2L starts 6,486 × 10% × 95% = 616; total 1,176 new patients → 588 on therapy on average → 1,176 × 6 × 0.95 × $18,500 × 0.76 = $94M.

Part 4 · The assumption book

Every input gets one row: ID, assumption, unit, low, base, high, source type, source, reason. Add an owner and a date when you version it. The forecast review is a review of this table.

ID	Assumption	Low	Base	High	Source
E0	Incident DLBCL cases, 2016–2030	×0.95	dlbcl_epi_inputs.csv	×1.05	dlbcl_epi_inputs.csv (sum of age bands); Teras et al. 2016 for the age pattern
E1	Incident-case growth after 2030	2%	2%	3%	2026–2030 growth in dlbcl_epi_inputs.csv
E2	Diagnosed and presenting	95%	97%	98%	dlbcl_treatment_funnel.csv, age-weighted
T1	Treated in 1L	82%	86%	89%	dlbcl_treatment_funnel.csv, age-weighted
T2	Progress to 2L (lifetime), one year later	26%	31%	36%	Week 2 cohort, Kaplan–Meier 36% at 36 mo; Sehn & Salles 2021 (>60% cured with R-CHOP)
T3	Progress to 3L (lifetime), one year later	43%	50%	57%	Week 2 cohort, Kaplan–Meier 61% at 24 mo
T4	Later-lines multiplier (3L+ starts per 3L patient)	1.14	1.24	1.34	dlbcl_treatment_funnel.csv
T5	Lag between lines	1	1	1	Week 2 cohort: median 14 mo (1L→2L), 9 mo (2L→3L)
B1	3L+ launch year	2,027	2,027	2,027	PDUFA January 2027 (fictional)
B2	2L label — first full year	2,030	2,029	2,028	Phase 3 readout; sBLA mid-2028 (fictional)
B3	Peak share of 3L+ starts	10%	15%	20%	Kalyanaram, Robinson & Urban 1995: share ≈ pioneer ÷ √order → 0.26 × 0.58
B4	Peak share of 2L starts	6%	10%	14%	2L crowded: CAR-T, ADC combination, chemo-immunotherapy
U1	Uptake: year 1 (% of peak share)	15%	30%	37%	analog_launches_monthly.csv: outpatient analogs 0.24–0.37; CAR-T 0.08–0.12
U2	Uptake: year 2	60%	90%	95%	same: outpatient 0.89–0.99; CAR-T 0.49–0.63
U3	Uptake: year 3	95%	1	1	same: all analogs 0.90–1.06
U4	Uptake: year 4 onward	1	1	1	same
C1	Coverage in launch year	55%	70%	85%	dlbcl_prescriber_3Lplus_starts.csv: top 10% of prescribers

					write 53% of starts
C2	Coverage from year 2	90%	95%	97%	Long tail of community prescribers
D1	Average months on therapy	5	6	7.5	8-cycle fixed course (label, fictional); persistence KM: analogs 5.9–9.4 mo; Cramer et al. 2008 definitions
D2	Compliance (doses received)	92%	95%	97%	Dose delays and holds
P1	List price (WAC) per 28-day month	18,500	18,500	18,500	Fictional
P2	Gross-to-net (share of WAC kept)	72%	76%	80%	Rebates, 340B, Medicare
L1	Loss-of-exclusivity year	2,038	2,039	2,040	Fictional; see IQVIA ADC paper for how exclusivity dates are read
L2	Share retained in the LOE year	65%	75%	85%	Mid-year entry
L3	Share retained after the LOE year	25%	40%	55%	Grabowski et al. 2021: small-molecule brands keep ~23% of units at 1 yr; biologics erode slower
F1	Launch-year phasing of starts		linear ramp (month m ∝ m)		
F2	Phasing after the launch year		flat (1/12 per month)		

Five source types. Data (claims, market, analogs) · Literature (peer-reviewed) · Analogs (similar launches) · Plans and labels (dates, price, dosing) · Judgment (written down, with a reason). Judgment is allowed; undocumented judgment is not.

Part 5 · Using data to justify an assumption

The data do not drive a complicated model; they justify a number in the book.

Assumption	Data and method	Result → book
T2, T3 progression	Week 2 claims cohort; Kaplan–Meier on time to next line	36% reach 2L (naive count 28%) → T2 = 31% after trimming for frail patients
D1 months on therapy	Kaplan–Meier on months on therapy for three analog regimens	means 5.9–9.4 months; label caps at 8 cycles → 6 months
U1–U3 uptake	share of peak reached in years 1–3 by analog launches	outpatient 24–37% in year 1; CAR-T 8–12% → 30% / 90% / 100%
B3 peak share	order-of-entry rule: share ≈ pioneer ÷ √order	0.26 ÷ √3 = 0.15
C1 coverage	prescriber concentration: top 10% write 53% of starts	formulary at the largest centers → 70% reachable in year 1

Censoring. A patient who has not relapsed by the end of the data is ‘not yet’, not ‘never’. Counting her as ‘never’ biases progression rates and months on therapy low. Kaplan–Meier fixes this: at each event time, the share still event-free is multiplied by (1 − events ÷ patients still at risk).

New patients vs patients on therapy. New patients are a flow per year; patients on therapy are a stock at a point in time. In a steady state, stock = flow × average time on therapy (Little's law). The field is paid on new patients; supply and revenue follow patients on therapy.

Part 6 · Ranges

Net revenue, $M	2027	2028	2029	2032	2035	Peak
Low	1	7	11	28	30	31
Base	9	37	52	94	101	108
High	37	148	228	251	274	308

Scenarios move every assumption to low (or high) together — the outer edges, deliberately extreme. **The tornado** moves one assumption at a time and ranks them by the swing they cause. Research the top bars first.

Tornado (2032, base $94M)	Low	High
B4 Peak share of 2L starts	-20	+20

D1 Average months on therapy	-16	+24
T2 Progress to 2L (lifetime), one year later	-15	+15
B3 Peak share of 3L+ starts	-15	+15
T3 Progress to 3L (lifetime), one year later	-6	+6
P2 Gross-to-net (share of WAC kept)	-5	+5

The plan conversation. The plan of record says $250M in 2029. That needs 3,119 new patients — 4.8× the base case and 85% of all 3L+ starts. Ask 'what would have to be true?' and put the answer in assumption terms.

Part 7 · Simple checks before the forecast leaves your desk

- **Back-cast.** Run the funnel on past years. Tonight it matches 2025 within 3% (1L +2%, 2L +1%, 3L+ −3%).
- **Market trend.** Exponential smoothing (ETS) on the monthly 3L+ starts: 4.6% error on the last 12 months vs 6.7% for 'same as last year'; trend and funnel agree within 2% for 2027.
- **Top-down split.** Last year's regional and account shares × the national forecast. Every piece adds back to the national number.
- **Loss of exclusivity.** Borrow other drugs' history: small-molecule brands keep about 23% of units one year after generic entry (Grabowski et al. 2021). Biologics erode slower — a documented judgment.
- **Prevalence model.** GLUCOREN 2025: 266M adults × 11.4% have T2D × 78% diagnosed × 72% treated × 47% add-on × 9.5% class × 4.7% brand × 8.8 fills = 313k prescriptions. Brand share is set so the model matches 2025.

Where time series and foundation models fit. Projecting a market or an in-line brand one or two years out. Not a launch: no history, no forecast. Foundation models (Chronos, TimesFM) need no fitting and help with many series at once; same limit.

Part 8 · Governance and pitfalls

IQVIA's study of ~12,000 launches: 50–70% miss their first-year forecast, usually on the high side. A written assumption book is how the bias gets caught.

- **Assumption book, versioned.** every input with a range, source, reason, owner, date
- **One model.** the budget is the first years of the LRP
- **Ranges every time.** scenarios and a tornado
- **Forecast vs actual.** track your own misses and their direction
- **Simple checks.** back-cast; trend; the pieces add up

Five common mistakes

- Undocumented assumptions — nobody can challenge them.
- Naive counts from censored data — 28% where the answer is 36%.
- Reading new patients as patients on therapy — persistence ignored.
- One number — no range, no tornado.
- A complicated model — nobody can follow it, so nobody trusts it.

Part 9 · Vocabulary → source

Term	Meaning tonight	Source
Budget forecast vs long-range plan	Two horizons from one model: 2–3 years by month; 5–15	Cook 2015

(LRP)	years by year	
Incidence vs prevalence	What the funnel starts from; prevalence ≈ incidence × duration	Centers for Disease Control and Prevention 2006
Flow vs stock model	Patients moving through time vs a yearly snapshot	Briggs 1998; Cook 2015
Assumption book	Every input with low / base / high, source and reason	Cook 2015; Gupta 2017
Censoring; Kaplan–Meier	Counting patients who have not had the event yet	Kaplan 1958; Clark 2003
Persistence; compliance	Months on therapy; doses taken as planned	Cramer 2008
New patients vs patients on therapy; Little's law	On therapy = new patients × months ÷ 12	Little 1961
Uptake ramp; analogs; Bass	Share of peak reached by year since launch	Bass 1969; IQVIA 2025
Order of entry; rank-size (Zipf)	Later entrants get less; a few prescribers write most	Kalyanaram 1995; Zipf 1949; Centers for Medicare & Medicaid Services
Loss of exclusivity	How much a brand keeps after generic entry	Grabowski 2021; IQVIA 2025
Exponential smoothing (ETS); top-down split	Market check; regional split	Hyndman 2021
Foundation models	Chronos, TimesFM, Lag-Llama, TimeGPT	Ansari 2024; Das 2024; Rasul 2023; Garza 2023
Launch forecast error	50–70% of launches miss their first-year forecast	Gupta 2017

Appendix · The data files (data/)

File	What it is
lymbrexa_assumptions.csv	the assumption book (27 rows)
dlbcl_epi_inputs.csv	population, incidence and new cases by age band, 2016–2030
dlbcl_treatment_funnel.csv	diagnosed, treated, progression and later-lines rates by age band
dlbcl_market_monthly.csv	national treatment starts by line, Jan 2019 – Jun 2026
dlbcl_market_by_region_account.csv	the same by region and account type
dlbcl_time_to_next_line.csv	months to next line, with a censoring flag (from the Week 2 cohort)
regimen_persistence_patients.csv	months on therapy, with a censoring flag, for three analog regimens
analog_launches_monthly.csv / _attributes.csv	share of line starts for 36 months after launch, eight analogs
dlbcl_prescriber_3Lplus_starts.csv	3L+ starts per prescriber, trailing 12 months
class_order_of_entry_shares.csv	share by order of entry in eight classes
glucoren_monthly_trx.csv; t2d_prevalence_funnel.csv; glucoren_assumptions.csv	the prevalence and LOE example
homework2/	the VELTRAXA / mantle cell lymphoma set for Homework 2

References

- Cook, A. G. **Forecasting for the Pharmaceutical Industry: Models for New Product and In-Market Forecasting and How to Use Them (2nd ed.).** Gower / Routledge, 2015. https://www.routledge.com/9781032838571
- Gupta, M. & McClellan, W. (IQVIA) **A Framework for Successful Biopharma Launches.** IQVIA white paper, 2017 — course library. Course library: a-framework-for-successful-biopharma-launches.pdf
- Centers for Disease Control and Prevention **Principles of Epidemiology in Public Health Practice (3rd ed.), Lesson 3: Measures of Risk.** CDC, 2006 (updated 2012). https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson3/section2.html
- Teras, L. R., DeSantis, C. E., Cerhan, J. R., et al. **2016 US lymphoid malignancy statistics by World Health Organization subtypes.** CA: A Cancer Journal for Clinicians 66(6):443–459, 2016. https://doi.org/10.3322/caac.21357
- IQVIA **Self-Driving CAR for Hematologic Malignancies.** IQVIA white paper, 2025 — course library. Course library: self-driving-car-for-hematologic-malignancies.pdf
- Sehn, L. H. & Salles, G. **Diffuse Large B-Cell Lymphoma.** New England Journal of Medicine 384:842–858, 2021. https://pubmed.ncbi.nlm.nih.gov/33657296/
- IQVIA **The Path to CAR T-cell Therapy: Barriers in LBCL.** IQVIA white paper, 2025 — course library. Course library: iqvia-the-path-to-car-tcell-therapy-white-paper-2025.pdf
- IQVIA **From Orphan to Opportunity: Mastering Rare Disease Launch Excellence.** IQVIA white paper, 2025 — course library. Course library: from-orphan-to-opportunity-mastering-rare-disease-launch-excellence.pdf

- Kaplan, E. L. & Meier, P. **Nonparametric Estimation from Incomplete Observations.** Journal of the American Statistical Association 53:457–481, 1958. https://doi.org/10.1080/01621459.1958.10501452
- Clark, T. G., Bradburn, M. J., Love, S. B. & Altman, D. G. **Survival Analysis Part I: Basic concepts and first analyses.** British Journal of Cancer 89:232–238, 2003. https://doi.org/10.1038/sj.bjc.6601118
- Briggs, A. & Sculpher, M. **An Introduction to Markov Modelling for Economic Evaluation.** PharmacoEconomics 13(4):397–409, 1998. https://doi.org/10.2165/00019053-199813040-00003
- Cramer, J. A., Roy, A., Burrell, A., et al. **Medication Compliance and Persistence: Terminology and Definitions.** Value in Health 11(1):44–47, 2008. https://doi.org/10.1111/j.1524-4733.2007.00213.x
- Little, J. D. C. **A Proof for the Queuing Formula: L = λW.** Operations Research 9(3):383–387, 1961. https://doi.org/10.1287/opre.9.3.383
- Bass, F. M. **A New Product Growth for Model Consumer Durables.** Management Science 15(5):215–227, 1969. https://doi.org/10.1287/mnsc.15.5.215
- Kalyanaram, G., Robinson, W. T. & Urban, G. L. **Order of Market Entry: Established Empirical Generalizations, Emerging Empirical Generalizations, and Future Research.** Marketing Science 14(3):G212–G221, 1995. https://doi.org/10.1287/mksc.14.3.G212
- Zipf, G. K. **Human Behavior and the Principle of Least Effort.** Addison-Wesley, 1949. https://archive.org/details/in.ernet.dli.2015.90211
- Centers for Medicare & Medicaid Services **Medicare Part D Prescribers — by Provider and Drug (public use file).** CMS, annual. https://data.cms.gov/provider-summary-by-type-of-service/medicare-part-d-prescribers
- Grabowski, H., Long, G., Mortimer, R. & Bilginsoy, M. **Continuing trends in U.S. brand-name and generic drug competition.** Journal of Medical Economics 24(1):908–917, 2021. https://pubmed.ncbi.nlm.nih.gov/34253119/
- IQVIA (Ark Patent Intelligence) **ADCs in Cancer.** IQVIA white paper, September 2025 — course library. Course library: iqvia-ark-pi-adc-cancer-whitepaper-09-25-forweb.pdf
- Hyndman, R. J. & Athanasopoulos, G. **Forecasting: Principles and Practice (3rd ed.).** OTexts, 2021 (free online). https://otexts.com/fpp3/
- Ansari, A. F., Stella, L., Turkmen, C., et al. (Amazon) **Chronos: Learning the Language of Time Series.** Transactions on Machine Learning Research, 2024. https://arxiv.org/abs/2403.07815
- Das, A., Kong, W., Sen, R. & Zhou, Y. (Google) **A decoder-only foundation model for time-series forecasting.** ICML 2024. https://arxiv.org/abs/2310.10688
- Rasul, K., Ashok, A., Williams, A. R., et al. **Lag-Llama: Towards Foundation Models for Probabilistic Time Series Forecasting.** arXiv preprint, 2023. https://arxiv.org/abs/2310.08278
- Garza, A., Challu, C. & Mergenthaler-Canseco, M. (Nixtla) **TimeGPT-1.** arXiv preprint, 2023 (vendor technical report). https://arxiv.org/abs/2310.03589